MILITARY SPECIFICATION

ANTENNA GROUP, OE-416(V)/SRC, 4-MHz TO 30-MHz RECEIVING TWIN WHIP

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, development, performance, and test requirements for the Antenna Group, OE-415(V)/SRC, hereinafter referred to as the antenna group. The antenna group is a nominal 4.1-meter (m) (14-foot (ft)) twin-whip configuration which provides broadband frequency coverage in the 4-megahertz (MHz) to 30-MHz range for high frequency (HF) shipboard receiving systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command, SPAWAR 003-121, Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5985

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SPECIFICATIONS

MILITARY

MIL-S-901	Shock Test, H.I. (High-Impact), Shipboard Machinery, Equipment And Systems, Requirements For
MIL-P-15024	Plates, Tags, And Bands For Identification Of Equipment
MIL-P-15024/5	Plates, Identification
MIL-E-16400G	Electronic, Interior Communication, And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts), Packaging Of

STANDARDS

MILITARY

MIL-STD-108	Definitions Of And Basic Requirement For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-965	Parts Control Program
MIL-STD-1310	Shipboard Bonding, Grounding, And Other Techniques For Electromagnetic Compatibility And Safety
MIL-STD-1364	Standard General Purpose Electronic Test Equipment

2.1.2 Government drawings. The following Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of the solicitation.

DRAWING

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

0330409	Installation Guidance Details, OE-( )/SRC Antenna (4-30 MHz, 14 ft) Twin Whip

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(Copies of specifications, standards, and drawings, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C95.2-66	Radio Frequency Radiation Hazard Warning Symbol
Y32.16-1975	Reference Designations For Electrical And Electronic Parts and Equipments

(Application for copies shall be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The antenna group shall be in accordance with MIL-E-16400G to the extent specified herein.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.1.2 Operational requirements. The antenna group shall give reliable performance within the limits specified herein, under conditions of intermittent or continuous operation during exposure to adverse conditions normally encountered in Naval shipboard communication service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance.

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3.1.3 Government-furnished material (GFM). GFM, if any, including either Government-furnished information (GFI) or Government-furnished equipment (GFE), shall be as specified (see 6.2 and 6.8).

3.2 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with the Parts, materials and processes paragraph of MIL-E-16400G to the extent specified herein.

3.2.1 Selection of parts, materials, and processes. Selection of parts, materials, and processes shall be in accordance with the Selection paragraph of MIL-E-16400G to the extent specified in 3.2.2

3.2.2 Parts. Parts shall be as specified in 3.2.2.1 through 3.2.2.4.

3.2.2.1 Individual selection. The performance of the antenna group shall not be dependent on the selection of individual parts, unless such selection is specifically approved by the procuring activity.

3.2.2.2 Nonstandard parts usage. Nonstandard parts shall be employed only with the approval of the procuring activity. Requests for approval to use nonstandard parts shall be submitted in accordance with MIL-STD-965.

3.2.2.3 Connectors, radio frequency (RF). RF connector and mating assemblies shall be as specified in SPAWAR Drawing (DWG) 0330409.

3.2.2.4 Insulators. Insulators shall be of a type to conform to the electrical requirements specified in 3.6.3. No coating material in any form shall be applied to any portion of ceramic insulation material that forms a part of the antenna group installation. Fiberglass insulators shall be treated to resist deterioration from sunlight and salt spray and shall be nonhygroscopic. The antenna insulator shall be a permanently assembled single part.

3.2.3 Materials. Materials shall be as specified in 3.2.3.1 through 3.2.3.4.

3.2.3.1 Selection of materials. Where metal-to-metal contact of dissimilar metals occurs, the selection of materials shall be in accordance with the Dissimilar metals paragraph of MIL-E-16400G, for combinations of metals subject to immersion, splashing, or spray from sea water.

3.2.3.1.1 Attachment hardware. Attachment hardware shall be limited to the types of metal-to-metal contact specified in a through d:

a. Aluminized hardware to aluminum metal
b. Brass hardware to brass metal
c. Brass hardware to austenitic corrosion-resistant metal, subject to the approval of the procuring activity
d. Corrosion-resistant steel-to-steel metal

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3.2.3.2 Unacceptable materials. Materials specified in the Unacceptable materials paragraph in MIL-E-16400G shall not be employed in the antenna group.

3.2.3.3 Antenna and coaxial cable termination box materials. Whip elements shall be made of spun aluminum. The coaxial cable termination box (feedbox) shall be made of aluminum. Whips shall be designed to mount on steel surfaces.

3.2.3.4 Magnetic materials. The antenna group shall have a minimum magnetic signature in accordance with the requirements specified in the Magnetic materials paragraph of MIL-E-16400G.

3.2.4 Processes. Processes shall be as specified in 3.2.4.1 and 3.2.4.2.

3.2.4.1 Soldering. Soldering shall conform to MIL-STD-454, Requirement 5.

3.2.4.2 Welds. Structural welding shall conform to MIL-STD-454, Requirement 13.

3.3 Design. Antenna group design shall be as specified in 3.3.1 through 3.3.9.1.

3.3.1 Antenna group capability. When installed in its design shipboard location, the antenna group shall provide a receiving capability in the 4-MHz to 30-MHz range. Major components of the antenna group shall be as specified in TABLE I.

TABLE I. Major components list.

Indenture	Item	Quantity	Nomenclature
1.0	Antenna Group	-	OE-416(V)/SRC
1.1	Antenna, 4.1-m (14-ft) whip	2	AS-3771(V)/U
1.2	Interconnecting feed assembly	1	
1.3	Coaxial cable termination box	1	J-( )/SRC
1.4	Mating connector assembly kit	1	--

3.3.2 Functional interface. The antenna group shall operate without tuning over the design frequency range and is intended for mounting on a tilting mechanism.

3.3.3 Internal interfaces. Internal interfaces of the antenna group shall include interconnections between the two whip antennas and the coaxial cable termination box, and between the coaxial cable termination box and the transmission line cable. The feedbox shall be located equidistant between the two whip antennas.

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3.3.4 Mechanical interfaces. The antenna group shall interface with an OA-( )(V)/SR tilting mechanism on the ship in accordance with the mounting configuration and requirements of SPAWAR DWG 0330409.

3.3.5 Protective devices. An electromagnetic pulse (EMP) protective device, Fischer Custom Communications Model FCC-250A-350A-IL-21/U, or equal, shall be installed in the coaxial cable termination box.

3.3.6 Reliability design. Reliability design shall be as specified in 3.3.6.1 and 3.3.6.2.

3.3.6.1 Derating. Temperature, power, and voltage stress levels and ratios, as applicable, shall be explicitly included in all mean-time-between-failures estimates. Derating shall be in accordance with MIL-STD-454, Requirement 18.

3.3.6.2 Tolerance sensitivity. The design of the antenna group shall minimize failures due to parameter change as a function of manufacturing tolerances, environmental factors, and age. The factors specified in a through h shall be considered:

a. Normal manufacturing tolerances, including soldering, wiring, placement, and machinery tolerances
b. Normal variations in material quality, including finish tolerances
c. Range of ambient temperatures and their rate of change
d. Vibration levels, steady-state and transient
e. Humidity
f. Operational aging, particularly that of dielectric materials from exposure to sun, ozone, and stack gasses
g. Nonoperational aging (storage life), shipping and handling effects
h. Salt spray and stack gas deposits, particularly those on insulators

3.3.7 Maintenance design. Maintenance design shall be as specified in 3.3.7.1 through 3.3.7.6.

3.3.7.1 Throw-away subassemblies. The designation of subassemblies as throw-away shall be subject to approval of the procuring activity.

3.3.7.2 Repairable subassemblies. Subassemblies designated as repairable shall be designed for ease of maintenance (see 6.6).

3.3.7.3 Adjustment and alignment. The antenna group shall not require component adjustment or alignment when installed in the ship, or during maintenance or repair.

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3.3.7.4 Accessibility. The coaxial cable termination box shall be so arranged that access to any part shall not require removal of other parts, except access covers.

3.3.7.5 Access covers. Access covers shall be retained by captive fasteners. Means shall be provided to prevent covers from falling or sliding away when opened.

3.3.7.6 Test equipment. The design of antenna group components shall be such that all maintenance and repair action may be performed using only standard general purpose electronic test equipment (GPETE), or substitute standard GPETE, as specified in MIL-STD-1364.

3.3.8 Mechanical design. The design of the antenna group shall conform to the Rounded corners and edges and Moisture pockets paragraphs in MIL-E-16400G.

3.3.8.1 Enclosures. The degree of enclosure for the feedbox shall be splashproof as specified in MIL-STD-108.

3.3.8.2 Antenna whip elements. The tops of the antenna whip elements shall be closed to prevent the entry of water and salt spray. The Antenna Assembly, AS-3771(V)/U, as a whole, shall be splashproof as specified in MIL-STD-108.

3.3.9 Safety design. The safety requirements for the antenna group shall be in accordance with MIL-STD-454, Requirement 1.

3.3.9.1 Electric shock hazard labeling. Hazard labels shall be permanently attached to each antenna in a position readily visible to personnel in the area and shall contain the following in the upper triangle:

DANGER
RADIO FREQUENCY SHOCK HAZARD

The following statement shall appear in the lower triangle:

A RADIO FREQUENCY SHOCK HAZARD MAY EXIST
TO PERSONNEL CONTACTING THE ANTENNA

Color and format shall conform to ANSI C95.2-66 requirements. Dimension D as specified in ANSI C95.2-66, Figure 1, shall be at least 0.3185 centimeter (cm) (0.125 inch (in.)).

3.4 Construction. Construction shall be as specified in 3.4.1 and 3.4.2.

3.4.1 EMP devices installation. The specified EMP devices (see 3.3.5) shall be mounted in the feedbox at the antenna feed point.

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3.4.2 Coaxial cable termination box. The coaxial cable termination box and all electrical connections shall be resistant to an increase in generation of intermodulation products as a result of the effects of salt air and constant vibration at 0.25 g amplitude. Techniques shall be employed such as, but not limited to, silver plating, brazing, or silver soldering.

3.5 Transportability. All antenna group equipment shall be transportable by air, rail, truck, barge, or ship. The maximum dimensions of any item, crated, shall not exceed 6.1 m (20 ft) in any dimension, nor 1.8 m (6 ft) in either of the other dimensions.

3.6 Performance characteristics. Performance characteristics shall be as specified in 3.6.1 through 3.6.3.

3.6.1 Frequency range. The antenna group shall conform to all the requirements of this specification over the frequency range of 4 MHz to 30 MHz without tuning.

3.6.2 Impedance and voltage standing wave ratio (VSWR). The impedance shall be 50 ohms nominal with a VSWR not exceeding 2:1. The impedance and VSWR shall be stable from unit to unit and before and after environmental tests.

3.6.3 Insulator voltage. Each whip insulator shall withstand the application of 60 kilovolts, direct current, when clean and dry, without arcing or visible corona.

3.7 Environmental requirements. Environmental requirements shall be as specified in 3.7.1 through 3.7.11.

3.7.1 Temperature, operating. The antenna group shall conform to all performance characteristics specified herein while operating at any temperature in the range -54° Celsius (C) to +65°C in accordance with Range 1 of MIL-E-16400G, for unsheltered equipment.

3.7.2 Temperature, nonoperating. The antenna group shall conform to all performance characteristics after long periods of nonoperation at ambient air temperatures in the range -62°C to +71°C in accordance with Range 1 of MIL-E-16400G, for unsheltered equipment.

3.7.3 Humidity. The antenna group shall have full operational capability when subjected to a relative humidity of 0 percent to 100 percent.

3.7.4 Salt fog (spray). The antenna group shall have full operational capability after being subjected to the test specified in 4.6.3.1.

3.7.5 Wind. The antenna group shall be capable of withstanding, without damage, winds having a relative velocity not less than 51.4 m per second (100 knots).

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3.7.6 Snow and ice loading. The antenna group shall be designed to withstand snow and ice loading on horizontal projected areas of 36.6 kilograms per square meter (kg/m^2) (7.5 pounds per square ft (lbs/ft^2)).

3.7.7 Icing. The antenna group, or portions thereof, exposed to the elements shall operate normally when covered with an ice load of 22.0 kg/m^2 (4.5 lbs/ft^2).

3.7.8 Shock. The antenna group shall conform to the requirements of MIL-S-901 for shock, Grade A, for deck-mounted equipment (see 6.5).

3.7.9 Vibration. The antenna group design criteria for environmental vibration shall be in accordance with MIL-STD-167-1, Type I.

3.7.10 Ship motion. Ship motion will be assumed to give the loading factors specified herein. These factors include gravity components as well as inertia effects. To obtain forces, multiply the factors by the weight of the structure. The antenna group shall be capable of operation without degradation of required performance under a downward load factor of 2.5 and a horizontal load factor of 1.2 regardless of equipment component or element location.

3.7.11 Nuclear air blast. The antenna group shall withstand a peak overpressure of 48.26 kilopascals (7 pound-force per square inch). Damage to components mounted in the coaxial cable termination box shall not be caused by deformation of the box.

3.8 Electromagnetic interference (EMI) suppression. EMI suppression shall be as specified in 3.8.1.

3.8.1 Bonding and grounding. The antenna group shall have provisions for post-installation bonding and grounding in accordance with the requirements of MIL-STD-1310.

3.9 Dimensions. The antenna group configuration shall be in accordance with the requirements of SPAWAR DWG 0330409.

3.10 Weight. The weight of the antenna group shall be not greater than 125 kilograms (275 pounds).

3.11 Color and finish. Painting of the feedbox and metal antenna elements shall conform to the requirements of the Painting paragraph of MIL-E-16400G, for the pretreating and painting of exposed (exterior use) metal enclosures and equipment. The third (final) coat shall be haze gray, Formula 151.

3.12 Identification plate or item markings. Identification plate or item markings shall be as specified in 3.12.1 through 3.12.4.

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3.12.1 Identification plates. Identification nameplates for the Antenna Group, AS-3771(V)/U and J-( )/SRC shall be affixed in conspicuous locations. Nameplate format, material, lettering application, and required information shall conform to MIL-P-15024 and MIL-P-15024/5.

3.12.2 Component reference designations. All parts and subassemblies shall be identified and marked. Identification and marking of elements shall be in accordance with the requirements of ANSI Y32.16-1975. Where it is not practicable to place the marking permanently on the element or adjacent structure, the element may be identified with temporary tags, sketches, or templates packed with the antenna group.

3.12.3 Information plates. Information plates shall comply with MIL-P-15024. Adhesive type plates may be used if installation methods and type are approved. Weatherproof plates and adhesives are required.

3.12.4 Insulator warning. Each whip insulator shall be indelibly marked on opposite sides (two places) with red block letters at least 2.54 cm (1 in.) high carrying the warning: DO NOT SAND OR PAINT.

3.13 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)
    1. In-process inspection (see 4.5.1)
    2. Production inspection (Group A) (see 4.5.2)
    3. Production control inspection (Group B) (see 4.5.3)
    4. Environmental inspection (Group C) (see 4.5.4)
c. Inspection of packaging (see 4.7)

4.3 Inspection conditions. Except where environmental tests require departure from these requirements, standard test conditions for all other inspections shall be those specified in the Standard test conditions paragraph of MIL-E-16400G.

4.4 First article inspection. Unless otherwise specified (see 6.2), two units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.5.1 through 4.5.6.

4.5.1 In-process inspection. In-process inspection shall be conducted on every antenna group offered for delivery. In-process inspection shall be performed during, or, where a process is performed on identical or representative samples, prior to the manufacture of an item. This inspection shall be comprised of examinations, tests, and demonstrations to prove the quality of workmanship and to verify adherence to specifications; for example, tests and demonstrations to detect departure from prescribed processes. In-process inspections shall include the examinations and tests of TABLE II.

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TABLE II. Examination and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection			
				In-process inspection	Group A	Group B	Group C
Parts, materials, and processes	3.2	4.6.1	X	X			
Internal interfaces	3.3.3	4.6.1	X				
Mechanical interfaces	3.3.4	4.6.1	X			X	
Protective devices	3.3.5	4.6.2	X	X			
Throw-away subassemblies	3.3.7.1	4.6.1	X				
Repairable subassemblies	3.3.7.2	4.6.1	X				
Alignment and adjustment	3.3.7.3	4.6.1	X				
Accessibility	3.3.7.4	4.6.1	X				
Access covers	3.3.7.5	4.6.1	X				
Mechanical design	3.3.8	4.6.1	X				
Antenna whip elements	3.3.8.2	4.6.1	X				
Safety design	3.3.9	4.6.1	X				
EMP device installation	3.4.1	4.6.1	X				
Coaxial cable termination box	3.4.2	4.6.1	X				
Transportability	3.5	4.6.1	X				
Bonding and grounding	3.8.1	4.6.1	X		X		
Dimensions	3.9	4.6.1	X		X		
Weight	3.10	4.6.1	X		X		
Identification plates	3.12.1	4.6.1	X	X			
Component reference designations	3.12.2	4.6.1	X	X			
Information plates	3.12.3	4.6.1	X	X			
Insulator warning	3.12.4	4.6.1	X	X			

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TABLE II. Examination and tests - Continued.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection			
				In-process inspection	Group A	Group B	Group C
Workmanship	3.13	4.6.1	X	X			
Frequency range	3.6.1	4.6.2	X				
Impedance and VSWR	3.6.2	4.6.2.1	X				
Insulator voltage	3.6.3	4.6.2	X				
Magnetic materials	3.2.3.4	4.6	X			X	
Welds	3.2.4.2	4.6 1/	X	X			
Enclosures	3.3.8.1	4.6	X				X
Antenna whip elements	3.3.8.2	4.6	X				X
Color and finish	3.11	4.6.1, 4.6.3.1	X	X			
Temperature, operating	3.7.1	4.6 2/	X				
Temperature, nonoperating	3.7.2	4.6 2/	X				
Humidity	3.7.3	4.6 2/	X				
Salt fog (spray)	3.2.3.1, 3.7.4	4.6.1, 4.6.3.1					
Wind	3.7.5	4.6	X				
Snow and ice loading	3.7.6	4.6	X				
Icing	3.7.7	4.6	X				
Shock	3.7.8	4.6.3.2	X				
Vibration	3.7.9	4.6.3.3	X				X
Ship motion	3.7.10	4.6	X				
Nuclear air blast	3.7.11	4.6	X				

1/ Testing shall be accomplished prior to fabrication of first article or production groups.
2/ Only an antenna base insulator shall be tested.

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4.5.2 Production inspection (Group A). Production inspection shall be made on every antenna group offered for delivery. Production inspection shall comprise examinations and tests to prove the quality of workmanship and to reveal omissions and errors of the production process; for example, tests to detect deviations from design and tests to detect hidden defects in materials. Production inspection shall include the examinations and tests in Group A of TABLE II.

4.5.3 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis. Production control inspection shall consist of examinations, tests, and analyses to determine performance throughout the required range of operation; to detect any deterioration of the design through wear of dies, molds, and jigs, and to reveal deviations in the processing of materials. Production control inspection shall be conducted on antenna groups which have passed the in-process and production inspections of 4.5.1 and 4.5.2, and shall consist of the examinations and tests specified for Group B of TABLE II. The production control inspection shall be performed on the complete antenna group. Unless otherwise specified (see 6.2), the sample for production control inspection shall be one group, or designated portions thereof, out of quantities of one to seven, and one additional group, or portion thereof, for each additional quantity of five, or fraction thereof.

4.5.4 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis. Environmental inspection shall encompass environmental tests to prove the durability of materials, parts, subassemblies and, as required, assemblies under exposure to the effects of vibration and enclosure tests. Environmental inspection shall include the examinations and tests specified in Group C of TABLE II. Unless otherwise specified (see 6.2), the basis for sampling shall be: when a first article test is conducted, no additional testing is required; when first article testing is waived in whole or in part, nonduplicative testing shall be performed on one antenna group that has completed Group B inspection.

4.5.5 Reinspection of conforming production control and environmental inspection samples. Unless otherwise specified (see 6.2), samples which have been subjected to, and have passed, production control or environmental inspection, or both, may be accepted on the contract provided that they are resubjected to, and pass, production inspection for workmanship after repair of all defects and damage incident to the Group B or Group C inspections.

4.5.6 Nonconforming production control and environmental sample units. If a sample unit fails inspection, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.4).

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4.6 Test methods. The examinations and tests specified in TABLE II shall be conducted in accordance with the applicable test methods of MIL-E-16400G, except as specified in 4.6.1 through 4.6.3.3.

4.6.1 Surface examinations. The antenna group shall be examined to the extent necessary to determine compliance with the applicable requirements of Section 3. Examinations shall employ qualitative inspections and quantitative measurements as appropriate.

4.6.2 Performance verification. Conformance to the requirements for performance specified in 3.6 shall be verified by measurement to demonstrate compliance.

4.6.2.1 Impedance and VSWR tests. Prior to and after the environmental testing specified in 4.6.3, the fully assembled antenna assembly shall be tested to measure impedance and VSWR. The measurements shall be made over a conducting ground screen consisting of at least 30 copper radials of No. 10 American Wire Gage. Each radial shall be at least 0.20 wavelengths long at the lowest frequency of the antenna to be measured. No vertically polarized conducting structure shall subtend an angle greater than 10 degrees within a 309.5-m (1000-ft) radius of the antenna feed point. The impedance shall be measured and expressed in terms of real and imaginary values ($R \pm jX$), and the VSWR shall be calculated, both at 1-MHz increments, including the low frequency and HF limits. Standard temperature, pressure, and humidity conditions are waived.

4.6.3 Environmental tests. Environmental tests shall be performed as specified in 4.6.3.1 through 4.6.3.3 and in accordance with the Performance of test paragraphs of MIL-E-16400G, except that operation during testing is not required. Upon completion of all environmental tests, the measurements of 4.6.2.1 shall be repeated. No measurement shall change more than 10 percent from the corresponding previous measurement.

4.6.3.1 Salt fog test. Tests of the antenna group shall be conducted in accordance with the Salt fog test paragraph of MIL-E-16400G, for exposed equipment, Procedure II. Instead of a fully assembled group, samples representative of every material and junctions of each combination of dissimilar materials shall be tested. The entire base insulator and the feedbox shall be tested. The salt fog test will provide testing to verify compliance with the painting processes of 3.11. The criteria of the Adhesion and blister resistance paragraph of MIL-E-16400G shall be used for evaluation of the painting processes.

4.6.3.2 Shock test. The antenna group shall be tested to determine conformance to the requirements of 3.7.8. The AS-3771(V)/U and J-( )/SRC may be tested individually.

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4.6.3.3 Vibration test. The antenna group shall be tested to determine conformance to the requirements of 3.7.9. One fully assembled antenna, AS-3771(V)/U, and the coaxial cable termination box, J-( )/SRC, with connecting feed wire shall be tested, spaced as intended in the installed configuration. The mounting fixture shall simulate a normal shipboard foundation. The maximum exciting frequency shall be not less than 33 hertz.

4.7 Inspection of packaging. Inspection shall be performed to ensure compliance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products acquired by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The antenna group covered by this specification is intended for shipboard use as a part of the communications antenna groups utilized in the DDG-51 Class guided missile destroyer.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. GFM, if any (see 3.1.3)
c. Number of first article samples to be submitted if other than specified in 4.4
d. If sampling for Group B inspection is to be other than specified in 4.5.3
e. If the sampling and the equipment for Group C inspection is to be other than as specified in 4.5.4
f. When reinspected conforming production control and environmental sample units may not be accepted (see 4.5.5)
g. Applicable levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article inspection is required, the item(s) should be a first production item. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for

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bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired and tested by the Government, and which has demonstrated acceptable quality and reliability. Bidders offering such products, who wish to rely on such production and test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Production control and environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.6).

6.5 Shock, Grade A. Shock, Grade A items are defined as: machinery, equipment, and systems essential for the safety and continued combat capability of the ship. Shock testing, or design of Grade A items to demonstrate that the item will continue to perform its principal function without significant change in performance, and that no portion of the equipment will come adrift or otherwise become a hazard to personnel or to Grade A machinery, equipment, or systems during and following the application of shock loadings.

6.6 Personnel skills. In designing equipment for ease of maintenance, information concerning Naval personnel skills and skill levels is available in NAVPERS 18068 and NAVPERS 15105. Rates of the Electronics Technician or Radioman ratings to be considered in performing maintenance tasks are Petty Officer Third Class (E-4) and Seaman Striker (E-3) (Class A school graduate).

6.7 Shipyard installation. The building shipyard will provide structural foundations suitable for the twin whip configuration of the antenna group, and conforming to SPAWAR DWG 0330409.

6.8 GFM, GFI, and GFE. The contracting officer should arrange to furnish the equipment specified in 3.1.3.

6.9 Subject term (key word) listing.

Antenna, 4.1-m (14-ft) Whip
Antenna Group, OE-416(V)/SRC
AS-3771(V)/U, Antenna, 4.1-m (14-ft) Whip
Coaxial Cable Termination Box, J-( )/SRC
Intended use: Shipboard, DDG 51 Class guided missile destroyers
J-( )/SRC, Coaxial Cable Termination Box
OE-416(V)/SRC, Antenna Group

6.10 Supersession data. This specification includes the requirements of SPAWAR-A-700A, dated 28 December 1987.

Preparing activity:
NAVY - EC

(Project No. 5985-N565-4(EC))

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-29520	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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